

ECE 410 Codefest 5 CMAN

1.
2 1 → ⁰⁰ 5 ⁰¹ 6

4 3 → ¹⁰ 7 ¹¹ 8

2.

Out C	Cycle	Input to row 0	Input to row 1	PE _{[0][0]}	PE _{[0][1]}	PE _{[1][0]}	PE _{[1][1]}
N/A	1	1	none	5	0	0	0
N/A	2	2	3	10	6	26	9
E[26,0]	3	none	4	0	12	38	30
E[0,0] E[26,30] E[38,0]	4	none	none	0	0	0	44

3. a) Total MAC = $5 \cdot 2^2$ (1 MAC operation per PE per cycle)
= 25 MAC operations

b) Each input value reused 1 time

c) # access A = 4
Access B = 4
Access C = 5

4. The outputs, C_{0-30} would remain fixed.